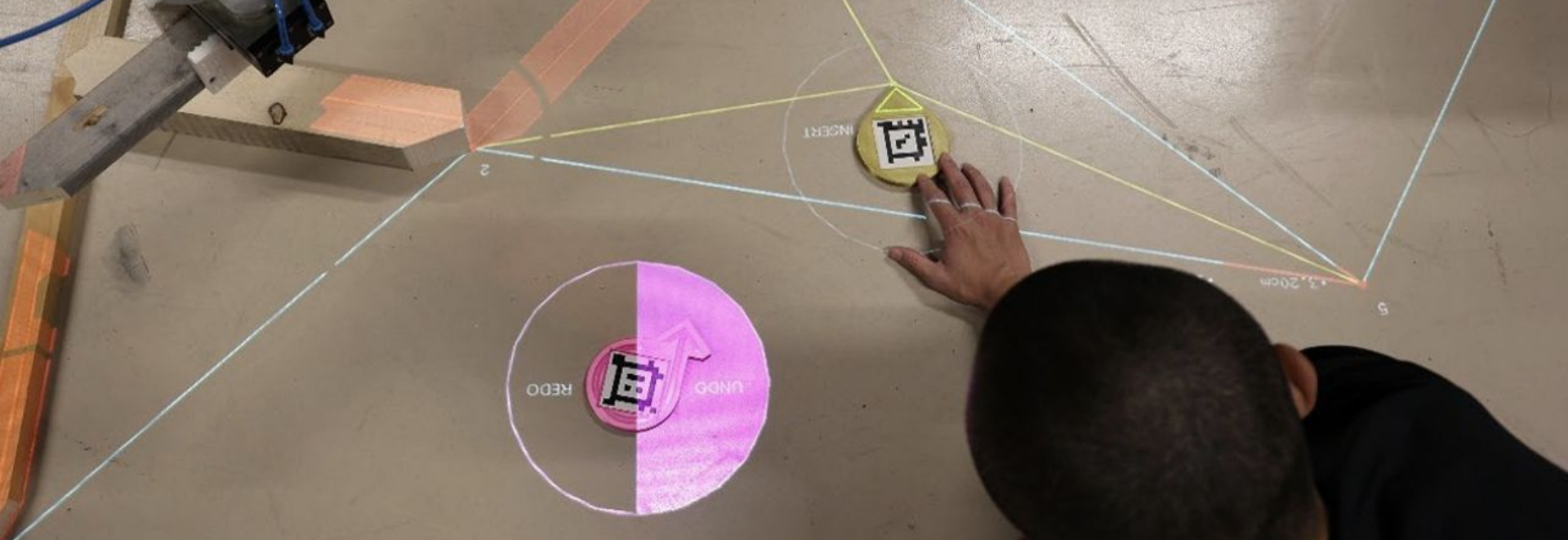




# Iaac

Master in Robotics and Advanced Construction  
Master in AI for Architecture and the Built Environment  
2025/2026

Hardware III  
Hamid Peiro, Aleksandra Kraeva

# Human-in-the-Loop

## Interactive Systems

This course is about designing systems where your body, decisions, and movements become the logic that drives a fabrication process.

Rather than programming a machine to execute a fixed output, you learn to design the *conversation* between human input and system behavior — using gesture, position, vision, and physical interfaces as your tools. The computational frameworks you'll work with (mapping strategies, finite state machines, behavioral logic) let you translate human action into meaningful, real-time responses.

The physical output of the course is a projection-mapped interactive system: light on a surface that reacts to what you do, guiding a fabrication process through rules you designed.

No touchscreens. No conventional apps. The interface is the space.

|           | Course 1<br>April 10                                    | Course 2<br>April 17                                   | Course 3<br>May 4            | Course 4<br>May 11                  | Course 5<br>May 18                     | Finals<br>May 22 |
|-----------|---------------------------------------------------------|--------------------------------------------------------|------------------------------|-------------------------------------|----------------------------------------|------------------|
| Morning   | Foundations:<br>Interaction,<br>AR & System<br>Thinking | Input &<br>Mapping<br>(From<br>Interaction to<br>Data) | FSM &<br>Rule-Based<br>Logic | Feedback &<br>Projection<br>Mapping | Integration &<br>Final<br>Presentation | FINALS           |
| Afternoon | Desk Crit<br>Q&A                                        | Desk Crit<br>Q&A                                       | Desk Crit<br>Q&A             | Desk Crit<br>Q&A                    | Desk Crit<br>Q&A                       |                  |

# Finals

## Group Project

From the first session, you **choose one modular assembly project and develop it across the entire course** — one layer at a time.

**Each session adds a new dimension to the same system:** first you define the assembly logic and rules, then you map human inputs to assembly behaviors, structure the sequence with a finite state machine, introduce real-time user control, and finally close the loop with projection feedback that tells you — live — what's valid, what's next, and what's wrong.

**By the final session, you have a working interactive prototype:** a physical assembly process guided by rules you designed, inputs you chose, and light that responds to both.

**Session 1 deliverable:** a short written proposal defining your module type, connection logic, and intended workflow. This becomes the foundation everything else is built on.



Hamid Peiro



Aleksandra  
Kraeva (Sasha)

# Instructions

Go to

**[www.menti.com](https://www.menti.com)**

Enter the code

**2265 9328**



Or use QR code

# Table of Content

**What is human interaction**

**Introduction to Interactive systems**

**UX in Spatial Systems**

**AR / Projection Mapping**

**Interaction & Fabrication & System Logic**

**Why this approach?**

---

# 01

## What is human interaction







# Three paradigms of making

## Traditional Craft

- Human hand decides
- Every action in real time
- Rich judgement
- Not scalable

## Digital Fabrication

- Front-loaded decisions
- Machine executes file
- Human steps back
- Deterministic output

## Human-in-the-Loop

- Decisions distributed
- Human steers the process
- System responds in real time
- Creative + computational

# Interaction is a conversation, not a command



## Command

You tell the machine exactly what to do. The machine executes.  
No conversation. No feedback loop. CAD → CNC is a command

---



## Interaction

You act. The system responds. You react to that response.  
You act again. The loop never fully closes — and that's the point.

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## Human-in-the-loop

Your judgment becomes part of the process logic.  
The system cannot proceed without your input at key decision points.

TOUCH  
MAGIX

CSML  
We make Con. a Habit.



# Types of human input

## Mouse / Click

Precise, learned, arbitrary — works on a flat plane only

## Touch / Pressure

More direct — still screen-bound, kills spatial awareness

## Gesture

Natural language of the body — wave, point, reach, rotate

## Body Position

Proximity, presence, orientation — no device required

## Shadow / Silhouette

The body as a mask — camera reads what light you block

## Vision / Gaze

Eye tracking as input — still emerging but powerful

## Physical Object

Placing a token, block, or material becomes a signal

## Sound / Voice

Amplitude, pitch, or command — spatial audio can be directional







IRoP uses a controller-based interaction system



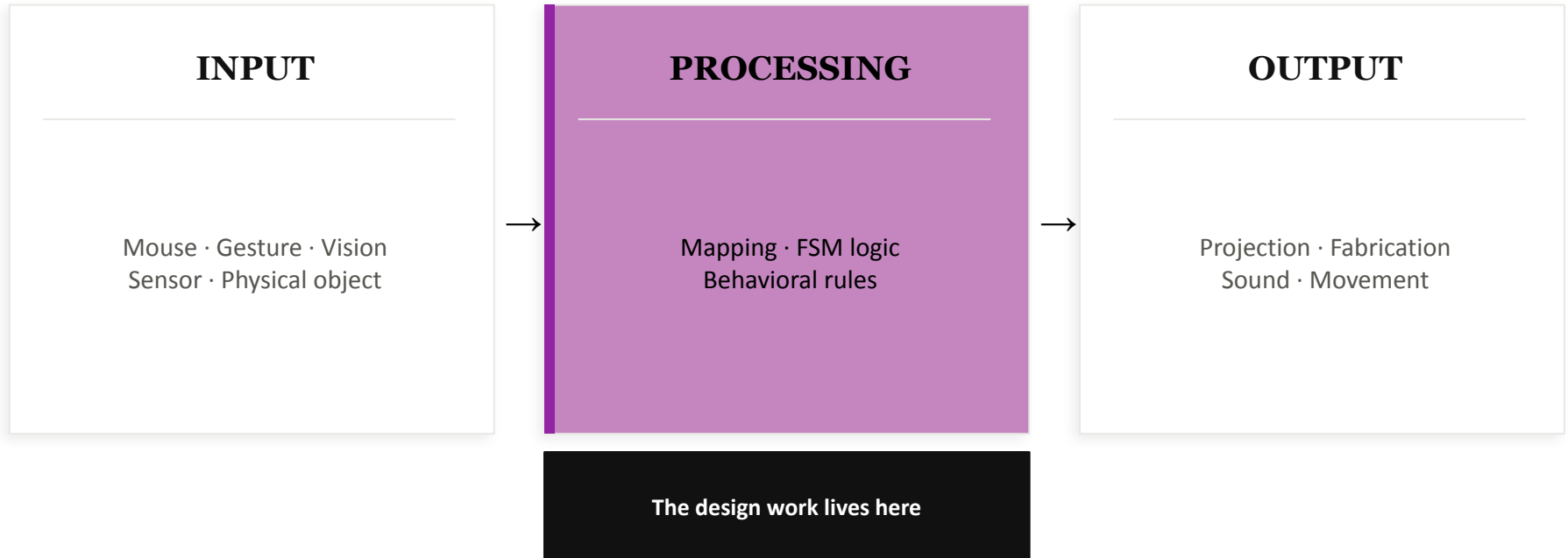


## 02

# Introduction to Interactive Systems



# Skeleton of any interactive system



# What makes it 'interactive' - *the feedback loop*

Feedback is not a feature. It is the mechanism that turns a sequence into a conversation.

## 01 Without feedback

Human acts. System responds. Human has no idea what happened. That's a command, not an interaction. Think: sending an email into a void.

## 02 Delayed feedback

System responds, but 2 seconds later. The human has already moved on. Feedback must be immediate to feel connected to the action that caused it.

## 03 Spatial feedback

The response happens in the same physical location as the action. This is what projection mapping enables — feedback that lives where the work is.



# Who is actually in control?

A webcam tracks your hand position over a table. As your hand moves closer to the edge, a projected fabrication path slows down and eventually stops. When you move back to center, it resumes at full speed.

**A**

The webcam — it sees everything and feeds the data

*The webcam is just a sensor. It has no opinion about what the data means.*

**B**

The person — they're always the one in control

*The person provides input — but without the logic, the machine doesn't know what to do with it.*

**C**

**The rule connecting hand position to fabrication speed**

*This is the design decision. The hardware is just hardware. The mapping rule is where the creativity lives.*

✓ correct

## 03

## UX in Spatial Systems







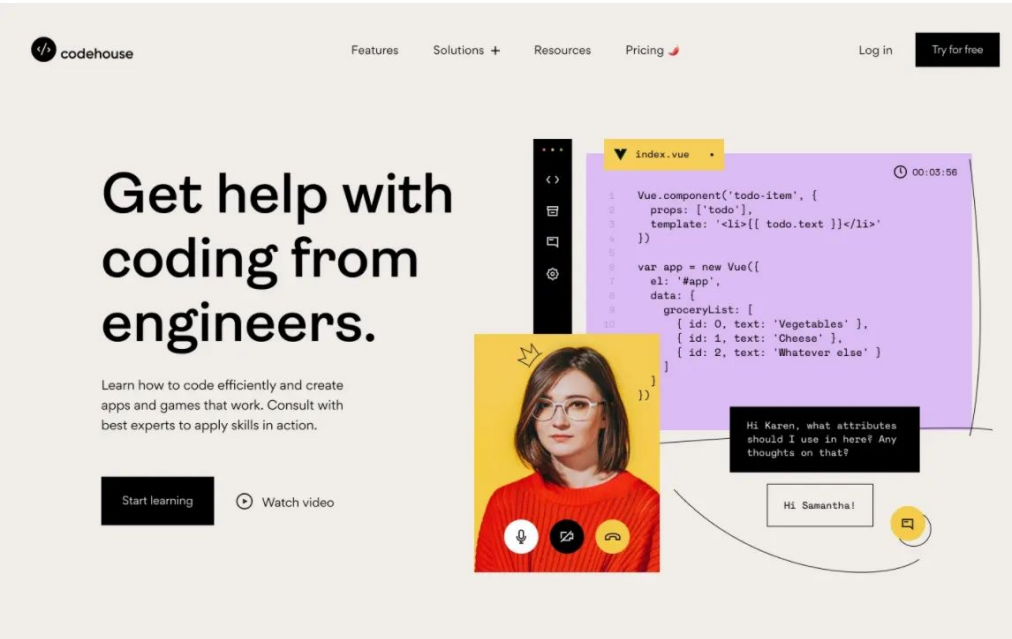
# Who is actually in control?

User Experience design originated in software. But the principles apply everywhere a human encounters a system.

- **Discoverability:** Can the user figure out what to do within 3 seconds? In spatial systems, discoverability is physical — does the space invite interaction?
- **Feedback:** Does the system acknowledge what the user did? Spatial feedback means the response happens where the action happened.
- **Error recovery:** When something goes wrong — wrong placement, wrong gesture — does the system help the user recover without frustration?
- **Cognitive load:** How much does the user have to remember or learn? Good spatial UX uses the body's existing vocabulary, not arbitrary new conventions.

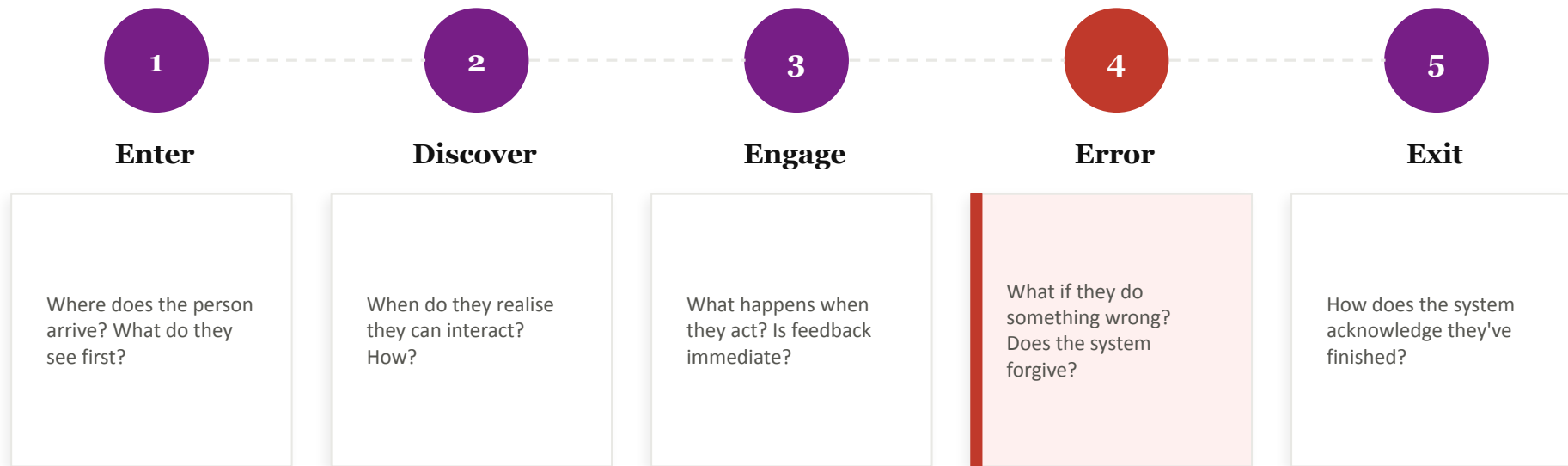


# Intuitive VS Confusing UX Design





# The user journey is a spa



***Your system must be readable within 3 seconds of first encounter — without anyone explaining anything.***

# What is Spatial UX Design?

Spatial UX is the discipline of designing user experiences where digital content exists in 3D physical space. This includes:

- **AR (Augmented Reality):** Overlays digital content on the real world.
- **VR (Virtual Reality):** Creates fully immersive digital environments.
- **MR (Mixed Reality):** Allows interaction between digital and physical elements.



Successful UX Design works without an explanation

# UX failures to avoid in your project

**01**

## The Punishment Error

Wrong placement resets everything. That is cruelty, not design. Good systems are forgiving: recover gracefully, help the user course-correct.

**04**

## The Touch Interface

A gestural system that behaves like a touchscreen is a missed opportunity. Avoid tap, swipe, pinch paradigms. Design for embodied space.

**02**

## The Delay Gap

More than 150ms between action and feedback and the brain disconnects the two. The causal link breaks. The system feels broken, not interactive.

**03**

## The Invisible State

The human doesn't know if the system saw them, processed their action, or is doing something. Anxiety kills engagement. Always show current state.

# Natural Interactions & Onboarding

01

## Mimic Real World

Use drag physics, shadow feedback, and realistic gestures. Integrate eye tracking, voice input, or hand gestures depending on the device.

02

## Smart Onboarding

**In AR:** Animate surface detection to visually reassure the user.

**In VR:** Use quick guided walkthroughs to teach gaze or gesture interactions.

01

## Spatial UIs should not mimic 2D screens. Instead:

Minimize flat panels.

Use curved UIs to match the user's visual arc.

Place interactive elements within a natural reach zone (roughly arm's length).

Embed UI into the spatial world rather than layering on top.

01

## Strategic UI Anchoring

There are two primary ways to position UI in AR/VR:

**World-Anchored UI:** Panels fixed in the environment (e.g., next to machines, on walls).

**Body-Anchored UI:** Menus on wrist, shoulder, or following gaze (e.g., wearable tools).

## 04

## AR / Projection Mapping





# Not a screen. A surface that knows you're there.



## **Not a video on a wall**

The content must respond to what the human is doing. Static projection is theatre, not interaction. The distinction matters.

---



## **Physical + digital merged — Spatial AR**

Projection mapping overlays computed imagery onto physical surfaces. AR without wearables. The world becomes the display.

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## **Light as fabrication instruction**

Where to place the next module. Which joint is valid. What state the system is in. All communicated through light on the work surface.

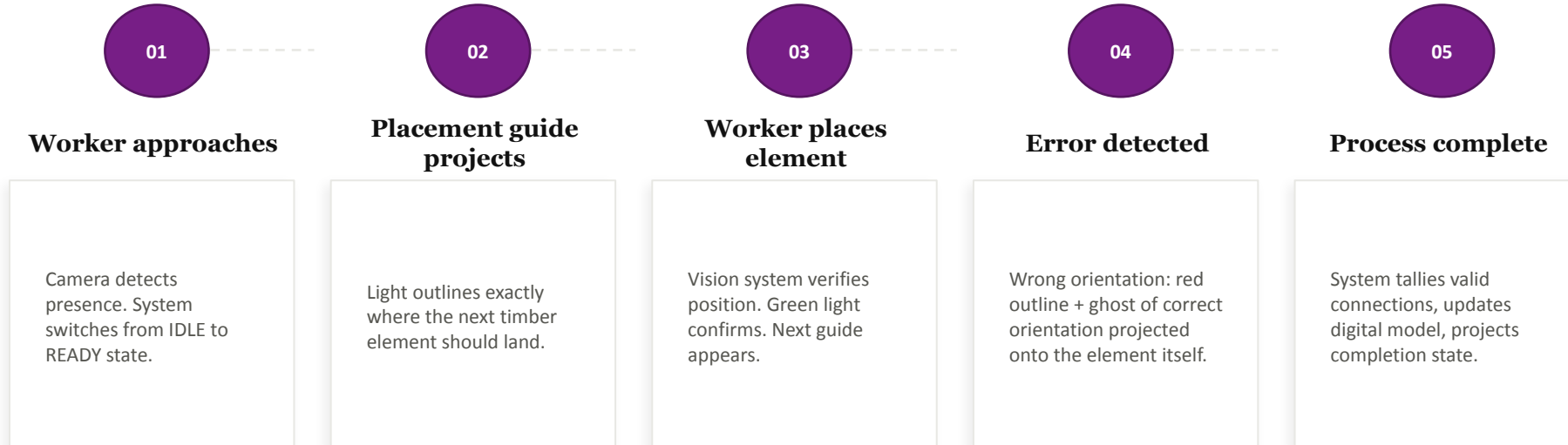
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## **The whole room is the interface**

Nobody wears anything. Nobody holds anything. The human moves freely. Presence, position, and gesture are the input.

# Projection as a fabrication guide - Example

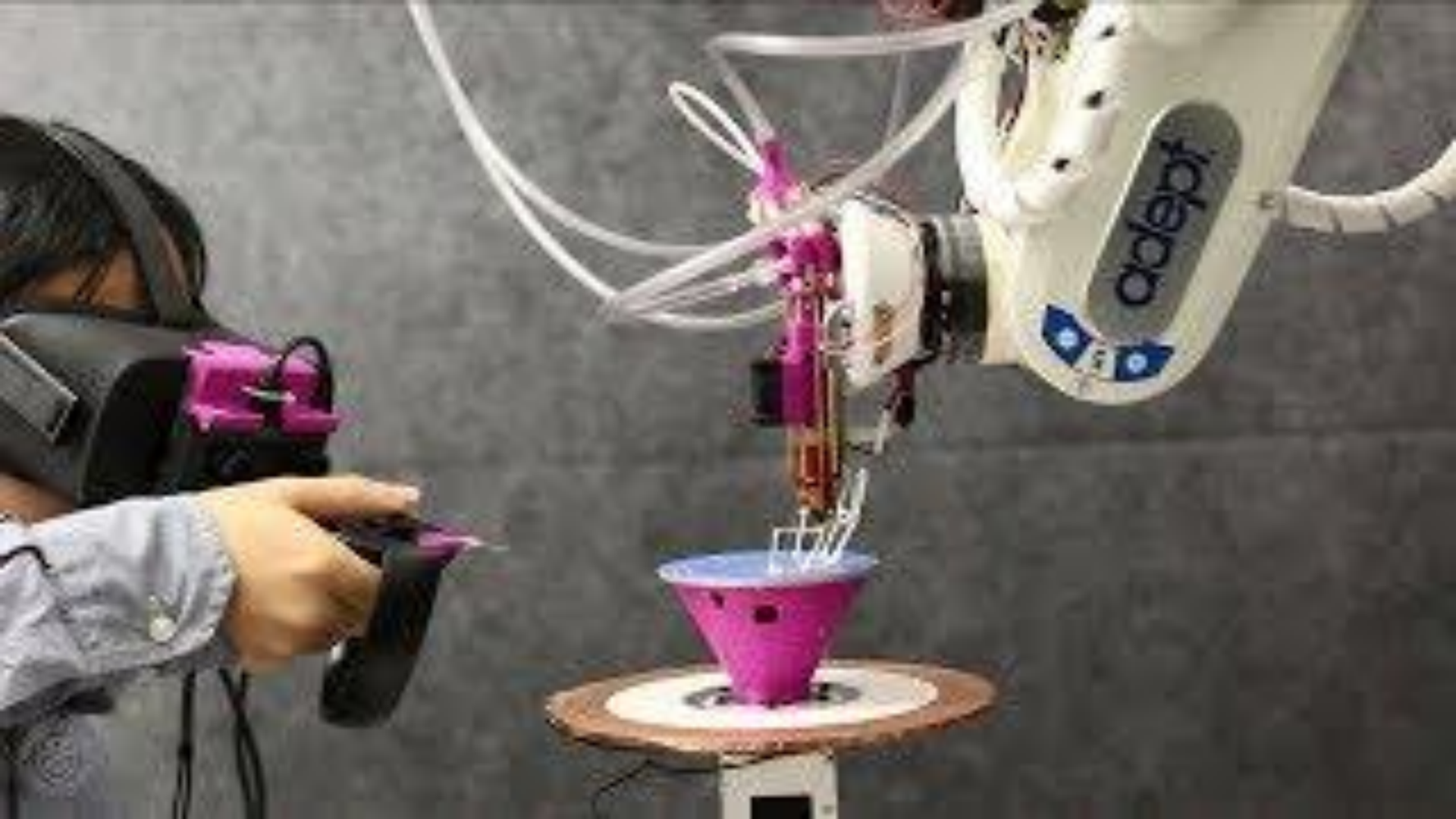


Augmented  
Reality



LEGO Augmented Reality Kit Build

D  
T  
Co.



# Your toolkit for this course

## Rhino + Grasshopper

Geometry, logic, parametric systems. Your main environment.

## Firefly (GH plugin)

Webcam input, Arduino sensors, real-time data into Grasshopper.

## Video Projector + HDMI

Basic HD resolution. €200 rental if needed. The output layer.

## Python in Grasshopper

Advanced logic, loops, FSM when Grasshopper components aren't enough.

## Anemone (GH plugin)

Loop logic and state machine patterns inside Grasshopper.

## TouchDesigner / HeavyM

For projection mapping UI and advanced visual feedback layers.

## Arduino / ESP32

Physical sensors — button, distance, pressure — as additional inputs.

## USB Webcam overhead

Body tracking, hand detection, shadow input. Feeds into Firefly.

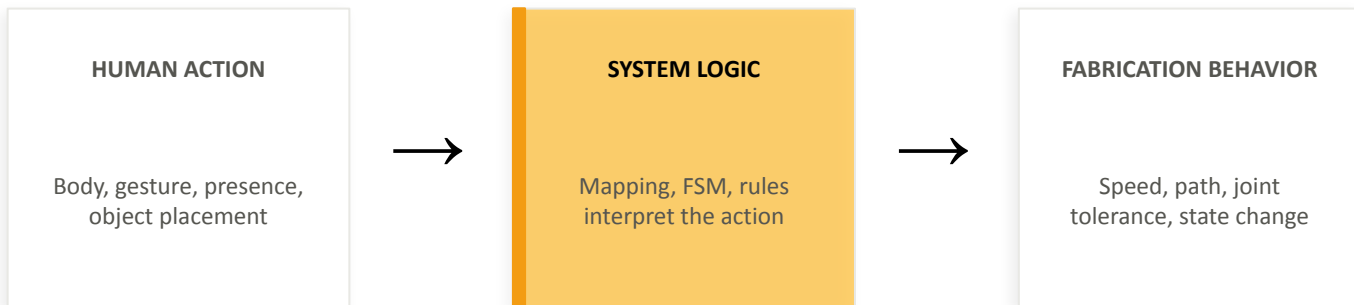
## 05

# Interaction & Fabrication & System Logic



# The chain: interaction - logic - fabrication

*These three things are not separate. In a well-designed system, they are one continuous chain. Your job is to design the connections.*



The creative design space is in the middle. Anyone can plug in a sensor. Anyone can drive a robot. The intelligence — the rules, the mapping, the state logic — that is what you will design.

# Mappin - the translation function

*Mapping connects an input value to an output behavior. It is not a technical detail — it is a design decision that shapes the entire feeling of an interaction.*

## LINEAR

### Direct

---

Every move produces equal change.  
Feels precise, neutral, professional.  
Good for calibration and measurement tasks.

## EXPONENTIAL

### Juicy

---

Small moves do little; large moves do a lot. Feels powerful and cinematic. Good for creative, expressive interactions.

## STEPPED

### Decisive

---

Output snaps to discrete values. Creates intentionality — the human feels like selecting, not sliding. Good for state-selection.



# Rules: logic between input and output

*Before you write a single line of code, you should be able to write your system's rules in plain language.*

The rule format:

**IF [condition] → THEN [action] → IN state [state name]**

- IF hand distance < 10cm → THEN scale projection zone → IN any state
- IF module position ≠ target → THEN project red outline → IN ACTIVE state
- IF no presence detected > 15s → THEN switch to ambient mode → IN ACTIVE state
- IF all modules valid → THEN trigger completion sequence → IN COMPLETE state

*Good rules are: specific, testable, and composable — they stack without contradiction.*

# Everything around you is in a state

## TRAFFIC LIGHT

STATE:

**RED**

Same input (push button)  
means different things in each  
state

## ELEVATOR

STATE:

**FLOOR 3  
DOOR CLOSED**

Button press only registers if  
not already at that floor

## VENDING MACHINE

STATE:

**WAITING  
FOR COIN**

'Give me product' input ignored  
until PAID state is reached

## YOUR PHONE

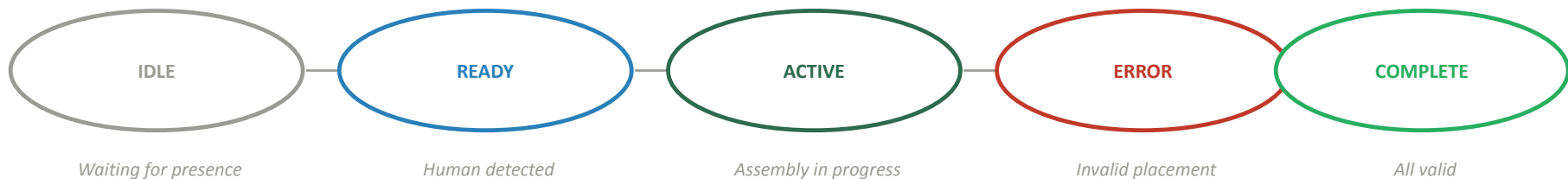
STATE:

**LOCKED**

Same swipe means unlock OR  
ignore, depending on current  
state

**A state is: what the system is doing RIGHT NOW — regardless of what it did before.**

# Finite State Machine: a map of all behaviours



- At any moment, the system is in exactly ONE state — never two simultaneously
- Same input can produce different output depending on the current state
- Transitions are triggered by events: a gesture, a sensor threshold, a timer expiring
- Draw the diagram before you write a single line of code — the diagram IS the design

# FSM in a fabrication context

*Let's design the FSM for a simple modular assembly system with projection feedback. What states does it need?*

**IDLE**

Ambient pattern. 'Ready when you are.'

→ Human enters sensing zone → READY

**READY**

Target zone highlighted. First placement guide.

→ Module placed → CHECKING

**CHECKING**

Blink while validating. 0.3 second max.

→ Valid → CONFIRMED | Invalid → ERROR

**CONFIRMED**

Green pulse. Next target appears.

→ Human lifts hand → READY for next

**ERROR**

Red outline + ghost of correct position.

→ Human corrects → CHECKING

**COMPLETE**

Full assembly celebration sequence.

→ All modules placed → COMPLETE

x4



Current Image  
(warpped by homography)



Estimated State  
and Action  
(action scaled x2)



Reference States  
for MPC



# How to draw your FSM

## 01 Define outputs per state

What does the projection do in each state? What does the system sound like? What does the fabrication process do?

## 02 Name the states

Short, uppercase, unambiguous. IDLE, SCANNING, PLACING, VALIDATING, ERROR, COMPLETE. One word or two.

## 03 Draw the transitions

For each state: what human action or sensor event causes a transition? Draw an arrow and label it with the trigger.

## 04 Find the gaps

Are there states you haven't designed for? What happens if the human is away for 60 seconds? What if power is interrupted?

---

# 06

## Why this approach?



# Screens put glass between the body and the world

01

## The body already knows

In architectural space, the body knows how to navigate, reach, and measure with gesture. We design systems that speak the body's language — not arbitrary conventions invented for a flat rectangle.

02

## Information in the right place

In cyber-physical workflows, feedback stays on the object being worked on — not on a screen the human must look away to read. The information should live where the action happens.

03

## Human judgment is the creative input

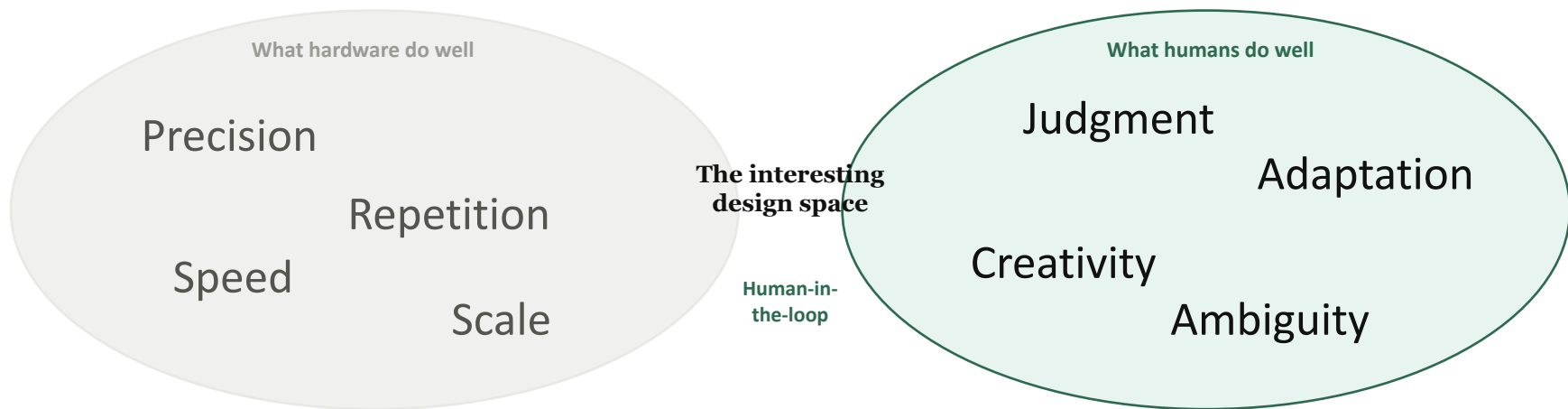
Robots execute fabrication. AI can generate form. But the intelligence that gives a process meaning — the judgment, the taste, the improvisation — still comes from the human in the loop.

*"The interface IS the space."*



# Hardware is an output Human is a collaborator

*Hardware appear in the output — they execute fabrication processes. But the intelligence that drives the robot does not come from an algorithm. It comes from the human in the loop.*



*You are designing the human's interface to the system — not the robot's programming. That interface IS the creative contribution.*

LIMELIGHT

# KICKIT TEAM TENNIS

LIGHT ART GAME





# Why we avoid touch screens

01

## Touch kills spatial awareness

The moment a human looks at a screen, they disconnect from the physical space. The work surface competes with the interface. In fabrication, that competition can cause errors.

02

## Touch is a learned convention

Swipe, tap, pinch — these are arbitrary conventions. They work for smartphones because we trained ourselves. But they're not natural. The body in space knows other things.

03

## Touch doesn't scale

A 2m assembly table cannot be a touchscreen. But it can have projection. The space itself can become the interface — no touch required.

04

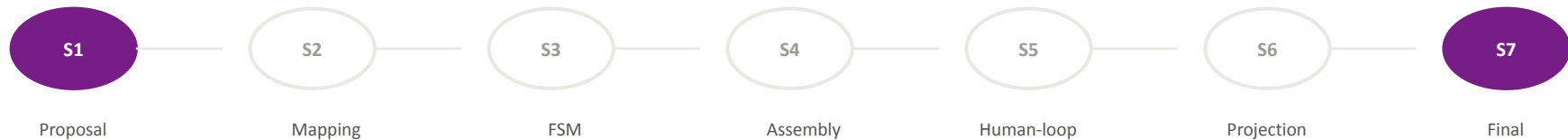
## Embodied interaction IS the research frontier

The entire field of HCI is moving toward spatial, gesture-based, and embodied interaction. This course puts you at the edge of that shift.

# One project for the whole term

## Make it smarter every session :)

You choose one modular assembly project at the start of this course and develop it across every session. By the end, you have a working interactive prototype — from concept to physical feedback system.





# What your proposal needs to answer

*Proposal format: max 1 page, or 3 slides. Due before Session 2.*

**What are the modules?**

*Physical units: bricks, struts, panels, meshes, tiles, wires. Define their geometry and material.*

**What is the connection logic?**

*How do modules connect? Snap fit, slot, gravity, adhesive, rope? What makes a connection valid vs invalid?*

**What human action drives the system?**

*Hand distance, body position, shadow, gesture, voice? Choose one input method to start with.*

**What does the projection feedback do?**

*Guide placement? Confirm validity? Show next step? Signal error? Be specific about what light communicates.*

**What is the final assembled thing?**

*A structure, a pattern, a functional object, a spatial installation, a game board?*

# Project management - MUST HAVE

Every team must submit and maintain a living project management schedule from Session 1 through to the final presentation. This is not a formality. It is evidence that you can structure your work.

We check and adjust it each session together, to maintain the progress and avoid frustration.

- 1) Submission each session
- 2) Submission for the finals (the whole roadmap with roles and tasks)

# Project management - MUST HAVE

Accepted formats (choose one or multiple, as you wish - minimum 1):

- Gantt chart — the industry standard. Use Notion, Excel, Google Sheets, or a dedicated tool like TeamGantt or Monday.com. Columns = weeks/sessions, rows = tasks, bars = duration, color = owner.
- Kanban board — Trello, Notion, or GitHub Projects. Three columns minimum: To Do / In Progress / Done. One card per task, one owner per card, one deadline per card.
- Sprint log — treat each session as a sprint. Define what you committed to before the session, deliver it, log what actually happened, adjust the next sprint. Works well in Notion or a shared Google Doc.



# Homework

REQUIRED

01

## Submit your project proposal

1 page or 3 slides. Module type, connection logic, intended human input, intended feedback method. Send before Session 2.

REQUIRED

02

## Project management Schedule

- 1) Detailed for the week before 2nd sessions (tasks, roles, dates, deliverables)
- 2) General “zoomed out” for the whole term
- 3) There should be an evidence that the whole team worked on the schedule (and project, obviously)

OBSERVE

03

## Find one embodied interaction in daily life

Not a touchscreen. Something where your body knows what to do without thinking. Photograph or sketch it. Share at the start of Session 2. It can be related to your final project or not

SKETCH

04

## Draw your first FSM

On paper. 3–5 states. Label the transitions. Don't worry about being right — worry about being specific. Bring it to Session 2.